

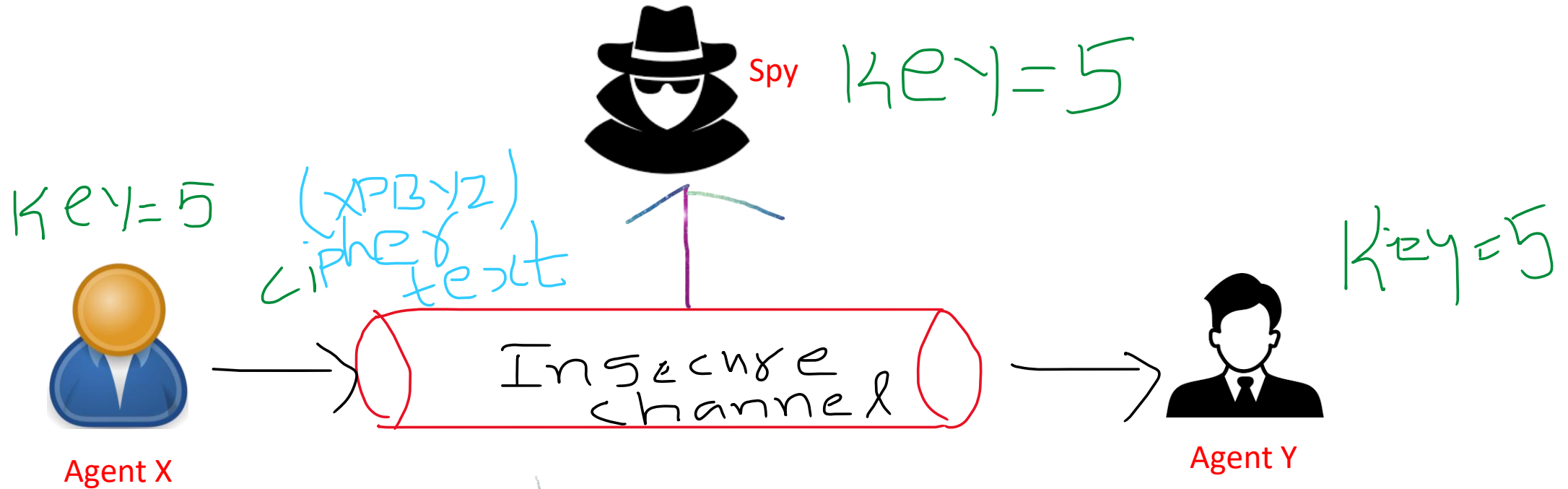
Diffie-Hellman Key Exchange & MITM Attack

Module III

Discrete logarithm problem

- $i = dlog_{b,m}(a)$
- $a = b^i \pmod{m}$
- we should get a unique value for i , if b is a primitive root of prime modulo m .
- $? = dlog_{17,2111}(1992)$
- $1922 = 17^i \pmod{2111}$
- Answer: 12

Symmetric key encryption: Big problem



key exchange problem?

Key Exchange Algorithm



Select $X_a < p$
 $X_a = 3$.

$X_a = 3$.
 $A = g^{X_a} \bmod p$
 $A = 2^3 \bmod 13$
 $A = 8$

Agent X



$S = B^{X_a} \bmod p$
 $S = 11^3 \bmod 13$
 $S = 5$

secret
key

$[P = 13, g = 2]$

p is prime number, g is generator.
 g is the primitive root of p

$A = 8$

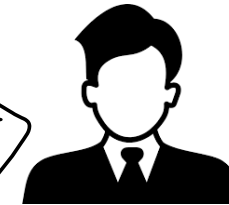
$B = 11$

public area

Select $Y_a < p$
 $Y_a = 7$.

$Y_a = 7$.
 $B = g^{Y_a} \bmod p$
 $B = 2^7 \bmod 13$
 $B = 11$

Agent Y



$S = A^{Y_a} \bmod p$
 $S = 8^7 \bmod 13$
 $S = 5$

secret
key

Key Exchange Algorithm



Spy

Can the hacker able to compute secret key by knowing p, g, A and B values. ?

Select $X_a < p$
 $X_a = 3$.

$X_a = 3$.
 $A = g^{X_a} \bmod p$
 $A = 2^3 \bmod 13$
 $A = 8$

Agent X



$S = B^{X_a} \bmod p$
 $S = 11^3 \bmod 13$
 $S = 5$

secret
key

$[P = 13, g = 2]$

p is prime number, g is generator.
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$A = 8$

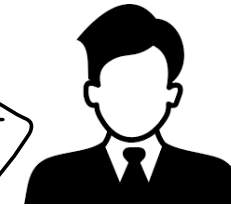
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Select $Y_a < p$
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Agent Y



$S = A^{Y_a} \bmod p$
 $S = 8^7 \bmod 13$
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secret
key

Diffie-Hellman Key Exchange

- Published in 1976 by Diffie and Hellman.
- It allows two parties who have not previously met to securely establish a key which they can use to secure their communications.
- The Diffie-Hellman key exchange was the first widely used method of safely developing and exchanging over an insecure channel.
- These keys can then be used with symmetric key algorithms to transmit Information in a protected manner.

Steps in Diffie-Hellman Key Exchange

- Agent X and Agent Y, using insecure communication, agree on a huge prime p and a generator g .
- They don't care if someone listens in.
- Agent X chooses some large random integer $X_a < p$ and keeps it secret.
- Likewise, Agent Y chooses $Y_a < p$ and keeps it secret.
- These are their private keys.
- Agent X computes his "public key" $A = g^{X_a} \bmod p$ and sends it to Agent Y using insecure communication.
- Agent Y computes its "public key" $B = g^{Y_a} \bmod p$ and sends it to Agent X. Here, $0 < A < p$, $0 < B < p$.
- Agent X computes $S1 = B^{X_a} \bmod p$ and Agent Y computes $S2 = A^{Y_a} \bmod p$.

